

# LITERATURE SURVEY REPORT

VL-JEPA vs. CLIP for CCTV Event Retrieval:  
Efficiency, Generalization, and Faithful Report Generation

*"From Footage to Facts."*

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# Abstract

This report presents a structured literature survey supporting the capstone project “VL-JEPA vs. CLIP for CCTV Event Retrieval: Efficiency, Generalization, and Faithful Report Generation.” The project investigates whether pretrained VL-JEPA (Joint Embedding Predictive Architecture for Vision-Language) representations can match or exceed CLIP-based representations for semantic event retrieval in CCTV surveillance footage, how well each representation generalizes across cameras and viewpoints under limited computational resources, and whether grounding LLM-generated incident reports in retrieved visual evidence reduces hallucinated content. To contextualize these questions, this survey reviews forty literature sources spanning eight thematic areas: (A) VL-JEPA foundations, (B) JEPA and self-supervised video representation learning, (C) CLIP and its retrieval-oriented derivatives, (D) CCTV and surveillance-specific vision-language systems, (E) video retrieval and retrieval-augmented generation (RAG), (F) detection, tracking and general video representation backbones, (G) LLM hallucination and faithfulness evaluation, and (H) cross-camera and cross-domain generalization. Each section synthesizes the reviewed works and identifies the specific gap that the proposed framework addresses.

## 1. Introduction

The proliferation of CCTV infrastructure in industrial, public, and residential settings has produced volumes of video data that are impractical to monitor or search manually. Vision-language models such as CLIP have made it possible to query video archives using natural language rather than manually engineered event categories, but CLIP was designed primarily for static image-text alignment and its behaviour on video, across changing camera viewpoints, and under constrained compute budgets is not fully characterised for the surveillance domain. VL-JEPA, a more recent joint-embedding predictive architecture, offers an alternative representation strategy that avoids autoregressive generation and has reported competitive results on general video-text benchmarks, but its suitability specifically for CCTV-style event retrieval remains under-explored. A second, largely independent problem arises once relevant footage has been retrieved: converting that evidence into a natural-language incident report using an LLM introduces a risk of hallucinated or unsupported claims, which is safety-critical in a surveillance context. This survey reviews the literature underpinning both halves of the problem — representation learning for retrieval, and faithfulness of generated reports — organised into the eight categories used for source collection.

### A. VL-JEPA Papers

VL-JEPA is the representation family at the centre of the project's main hypothesis (H1, H2). These two sources define the architecture and its officially reported evaluation, and form the baseline against which the project's own CCTV-domain experiments are compared.

| S.No | Paper Title                                                          | Year | Application                                                             | Link                                                                                                  |
|------|----------------------------------------------------------------------|------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 1    | VL-JEPA: Joint Embedding Predictive Architecture for Vision-Language | 2025 | Vision-language retrieval, video understanding, VQA, semantic retrieval | <a href="https://arxiv.org/abs/2512.10942">https://arxiv.org/abs/2512.10942</a>                       |
| 2    | VL-JEPA (ICLR / OpenReview Version)                                  | 2025 | Official conference version                                             | <a href="https://openreview.net/forum?id=tjimrqc2BUB">https://openreview.net/forum?id=tjimrqc2BUB</a> |

**Synthesis.** Together these establish VL-JEPA as a joint-embedding predictive approach to vision-language representation that, unlike CLIP-style contrastive image-text encoders, learns by predicting representations rather than raw pixels or tokens. The project treats the general-domain results reported here as the starting hypothesis to be

re-tested under CCTV-specific conditions: lower resolution, static viewpoints, and resource-constrained inference.

## B. JEPA / Video Representation Papers

This category covers the broader JEPA family and self-supervised video encoders that VL-JEPA builds on conceptually, providing background for objective 1 (framework design) and objective 4 (computational analysis).

| S.No | Paper Title                                                                          | Year | Application                                      | Link                                                                            |
|------|--------------------------------------------------------------------------------------|------|--------------------------------------------------|---------------------------------------------------------------------------------|
| 1    | V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning | 2025 | Video understanding, robotics, action prediction | <a href="https://arxiv.org/abs/2506.09985">https://arxiv.org/abs/2506.09985</a> |
| 2    | V-JEPA 2.1                                                                           | 2026 | Dense video representation learning              | <a href="https://arxiv.org/abs/2603.14482">https://arxiv.org/abs/2603.14482</a> |

**Synthesis.** V-JEPA and its successors demonstrate that predictive, non-generative self-supervised objectives can yield transferable video representations useful for downstream understanding and planning tasks. These works motivate treating VL-JEPA as a video-native alternative to image-pretrained CLIP encoders, and support the project's framing that predictive embeddings may generalize differently than contrastive ones when camera viewpoint or domain shifts, which is directly relevant to hypothesis H2.

## C. CLIP Papers

CLIP and its retrieval-oriented derivatives form the comparison baseline for objectives 1-3, since the project explicitly measures VL-JEPA against CLIP-family representations on retrieval accuracy, generalization, and efficiency.

| S.No | Paper Title                                                                  | Year | Application                            | Link                                                                            |
|------|------------------------------------------------------------------------------|------|----------------------------------------|---------------------------------------------------------------------------------|
| 1    | Learning Transferable Visual Models From Natural Language Supervision (CLIP) | 2021 | Semantic retrieval, zero-shot learning | <a href="https://arxiv.org/abs/2103.00020">https://arxiv.org/abs/2103.00020</a> |
| 2    | VideoCLIP: Contrastive Pre-training for Zero-shot Video-Text Understanding   | 2021 | Video retrieval, video captioning      | <a href="https://arxiv.org/abs/2109.14084">https://arxiv.org/abs/2109.14084</a> |
| 3    | VadCLIP: Adapting CLIP for Video Anomaly Detection                           | 2023 | CCTV anomaly detection                 | <a href="https://arxiv.org/abs/2307.13425">https://arxiv.org/abs/2307.13425</a> |
| 4    | CLIP-ReID                                                                    | 2022 | Cross-camera person retrieval          | <a href="https://arxiv.org/abs/2211.13977">https://arxiv.org/abs/2211.13977</a> |
| 5    | RegionCLIP                                                                   | 2022 | Open-vocabulary object detection       | <a href="https://arxiv.org/abs/2112.09106">https://arxiv.org/abs/2112.09106</a> |

**Synthesis.** The original CLIP paper established large-scale contrastive image-text pretraining as a route to zero-shot semantic retrieval, and the remaining four papers show the surveillance-relevant lineage that follows from it: VideoCLIP extends the contrastive objective to video-text pairs; VadCLIP and RegionCLIP adapt CLIP features to anomaly detection and region-level open-vocabulary detection respectively; and CLIP-ReID applies CLIP-derived embeddings to cross-camera person retrieval. This body of work confirms CLIP's flexibility across surveillance-adjacent tasks while also indicating that most gains come from task-specific adaptation rather than the base encoder alone — a gap the project addresses by testing pretrained (not fine-tuned) CLIP and VL-JEPA under matched, resource-constrained conditions.

## D. CCTV / Surveillance Papers

These sources anchor the project within current CCTV-specific vision-language systems, informing the problem statement and the report-generation component (objective 5).

| S.No | Paper Title                                                                                         | Year | Application                       | Link                                                                            |
|------|-----------------------------------------------------------------------------------------------------|------|-----------------------------------|---------------------------------------------------------------------------------|
| 1    | AssistPDA: Online Video Surveillance Assistant for Video Anomaly Prediction, Detection and Analysis | 2025 | AI surveillance assistant         | <a href="https://arxiv.org/abs/2503.21904">https://arxiv.org/abs/2503.21904</a> |
| 2    | Cerberus: Cascaded Vision-Language Models for Real-Time Video Anomaly Detection                     | 2025 | Real-time anomaly detection       | <a href="https://arxiv.org/abs/2510.16290">https://arxiv.org/abs/2510.16290</a> |
| 3    | VERA: Verbalized Learning for Explainable Video Anomaly Detection                                   | 2025 | Explainable surveillance          | <a href="https://openaccess.thecvf.com/">https://openaccess.thecvf.com/</a>     |
| 4    | The Evolution of Video Anomaly Detection: From DNN to MLLM                                          | 2025 | Survey of video anomaly detection | <a href="https://arxiv.org/abs/2507.21649">https://arxiv.org/abs/2507.21649</a> |

**Synthesis.** The surveyed systems trace a shift in CCTV analytics from single-purpose deep networks toward vision-language and multimodal-LLM-based assistants (AssistPDA, Cerberus) that can explain their decisions in natural language (VERA), a trajectory summarised by the DNN-to-MLLM survey. This confirms that natural-language grounded surveillance analysis is an active research direction, but also shows that few of these systems directly compare a joint-embedding predictive representation like VL-JEPA against CLIP under identical CCTV retrieval conditions — the specific gap this project targets.

## E. Video Retrieval & RAG Papers

These works inform the retrieval-augmented report generation pipeline used to satisfy objective 5 (faithfulness of generated incident reports).

| S.No | Paper Title                                                | Year | Application                         | Link                                                                            |
|------|------------------------------------------------------------|------|-------------------------------------|---------------------------------------------------------------------------------|
| 1    | VideoRAG: Retrieval-Augmented Generation over Video Corpus | 2025 | Video retrieval + report generation | <a href="https://arxiv.org/abs/2501.05874">https://arxiv.org/abs/2501.05874</a> |
| 2    | VideoRAG for Extremely Long Context Videos                 | 2025 | Long CCTV video search              | <a href="https://arxiv.org/abs/2502.01549">https://arxiv.org/abs/2502.01549</a> |
| 3    | MissionGNN                                                 | 2025 | Knowledge graph from events         | <a href="https://openaccess.thecvf.com/">https://openaccess.thecvf.com/</a>     |
| 4    | GenDFIR                                                    | 2024 | Timeline generation using RAG       | <a href="https://arxiv.org/abs/2409.02572">https://arxiv.org/abs/2409.02572</a> |

**Synthesis.** VideoRAG and its long-context variant show that retrieval-augmented generation can ground LLM outputs in retrieved video segments rather than free generation, directly relevant to reducing hallucination in CCTV incident reports. MissionGNN's knowledge-graph construction from events and GenDFIR's RAG-based timeline generation further indicate that structuring retrieved evidence (as events, graphs, or timelines) before text generation is a common strategy for improving faithfulness — a design principle the project adopts in evaluating report faithfulness against retrieved visual evidence (hypothesis H3).

## F. Detection, Tracking & Video Representation Papers

These establish the supporting computer-vision building blocks (detection, tracking, action and pose recognition) referenced under Subject Knowledge and Technical Requirements.

| S.No | Paper Title                             | Year | Application                    | Link                                                                            |
|------|-----------------------------------------|------|--------------------------------|---------------------------------------------------------------------------------|
| 1    | VideoMAE                                | 2022 | Self-supervised video learning | <a href="https://arxiv.org/abs/2203.12602">https://arxiv.org/abs/2203.12602</a> |
| 2    | SlowFast Networks for Video Recognition | 2019 | Action recognition             | <a href="https://arxiv.org/abs/1812.03982">https://arxiv.org/abs/1812.03982</a> |
| 3    | ByteTrack                               | 2021 | Multi-object tracking          | <a href="https://arxiv.org/abs/2110.06864">https://arxiv.org/abs/2110.06864</a> |
| 4    | DeepSORT                                | 2017 | Object tracking                | <a href="https://arxiv.org/abs/1703.07402">https://arxiv.org/abs/1703.07402</a> |
| 5    | ST-GCN                                  | 2018 | Human pose & fall detection    | <a href="https://arxiv.org/abs/1801.07455">https://arxiv.org/abs/1801.07455</a> |

**Synthesis.** VideoMAE and SlowFast represent masked-autoencoding and two-stream convolutional approaches to general video representation learning, respectively, and serve as older, non-vision-language baselines for comparison. ByteTrack and DeepSORT are established multi-object tracking algorithms that would be required if the framework is extended from clip-level retrieval to persistent object-level surveillance analytics, while ST-GCN's skeleton-based graph convolutional approach is relevant to pose- and fall-detection use cases. These are included as supporting infrastructure rather than as direct competitors to VL-JEPA or CLIP in the retrieval comparison.

## G. LLM Hallucination & Faithfulness Papers

These papers provide the evaluation methodology for hypothesis H3 — measuring whether grounding reduces hallucination in generated incident reports.

| S.No | Paper Title                                      | Year | Application                    | Link                                                                            |
|------|--------------------------------------------------|------|--------------------------------|---------------------------------------------------------------------------------|
| 1    | SelfCheckGPT                                     | 2023 | Hallucination detection        | <a href="https://arxiv.org/abs/2303.08896">https://arxiv.org/abs/2303.08896</a> |
| 2    | Hallucination in Large Language Models: A Survey | 2023 | LLM hallucination survey       | <a href="https://arxiv.org/abs/2311.05232">https://arxiv.org/abs/2311.05232</a> |
| 3    | Faithfulness in Natural Language Generation      | 2022 | Report faithfulness evaluation | <a href="https://arxiv.org/abs/2207.07408">https://arxiv.org/abs/2207.07408</a> |

**Synthesis.** SelfCheckGPT proposes a sampling-based, reference-free method for detecting hallucinated content in LLM outputs, while the accompanying survey categorises hallucination types and mitigation strategies across the broader LLM literature. The faithfulness paper addresses the related but distinct problem of evaluating whether generated text is supported by its source content in natural language generation tasks. Together, these give the project a methodological basis for scoring CCTV incident reports for unsupported claims relative to retrieved visual evidence, rather than relying on ad hoc manual review alone.

## H. Cross-Camera Generalization Papers

These works directly support objective 3 and hypothesis H2, which concern generalization of retrieval performance across cameras and viewpoints.

| S.No | Paper Title | Year | Application                                | Link                                                                            |
|------|-------------|------|--------------------------------------------|---------------------------------------------------------------------------------|
| 1    | CLIP-ReID   | 2022 | Cross-camera retrieval                     | <a href="https://arxiv.org/abs/2211.13977">https://arxiv.org/abs/2211.13977</a> |
| 2    | TransReID   | 2021 | Transformer-based person re-identification | <a href="https://arxiv.org/abs/2102.04378">https://arxiv.org/abs/2102.04378</a> |

| S.No | Paper Title                                          | Year | Application                  | Link                                                                            |
|------|------------------------------------------------------|------|------------------------------|---------------------------------------------------------------------------------|
| 3    | Deep Learning for Person Re-identification: A Survey | 2020 | Cross-camera matching survey | <a href="https://arxiv.org/abs/2001.04193">https://arxiv.org/abs/2001.04193</a> |

**Synthesis.** CLIP-ReID and TransReID represent two approaches — vision-language adaptation and pure transformer architectures, respectively — to the cross-camera person re-identification problem, where the same subject must be matched across non-overlapping camera views. The accompanying survey situates both within the broader re-identification literature. While this project's retrieval task (semantic event retrieval, not identity matching) differs from re-identification, the cross-camera generalization challenge is structurally similar, and these papers inform the evaluation protocol used to test VL-JEPA and CLIP across different cameras or video sources.

## 2. Cross-Cutting Discussion and Research Gap

Reviewing the eight categories together highlights a consistent pattern: VL-JEPA (Category A, B) and CLIP (Category C) have each been evaluated extensively on general-purpose video-text benchmarks and, in CLIP's case, adapted successfully to several surveillance-adjacent tasks such as anomaly detection (VadCLIP) and cross-camera retrieval (CLIP-ReID). Modern CCTV-specific systems (Category D) are increasingly built on vision-language and multimodal-LLM foundations, and video-RAG methods (Category E) show that retrieval grounding is an effective strategy for constraining LLM-generated text. Hallucination-evaluation methods (Category G) exist and are applicable to this grounding step. Detection and tracking baselines (Category F) and re-identification literature (Category H) provide supporting infrastructure and a template for cross-camera evaluation protocols.

No reviewed source, however, directly benchmarks pretrained VL-JEPA against pretrained CLIP specifically for CCTV-style event retrieval under resource-constrained inference, nor combines that comparison with an evaluation of hallucination in the resulting incident reports. This is the gap the capstone project is designed to fill: a lightweight, reproducible framework that (i) compares pretrained VL-JEPA and CLIP representations on CCTV retrieval metrics, (ii) tests generalization across cameras/viewpoints, (iii) profiles computational cost under constrained resources, and (iv) evaluates the faithfulness of LLM-generated incident reports grounded in the retrieved evidence.

## 3. Consolidated Reference List

### A. VL-JEPA Papers

- [1] VL-JEPA: Joint Embedding Predictive Architecture for Vision-Language (2025). <https://arxiv.org/abs/2512.10942>
- [2] VL-JEPA (ICLR / OpenReview Version) (2025). <https://openreview.net/forum?id=tjimrqc2BUB>

### B. JEPA / Video Representation Papers

- [3] V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning (2025). <https://arxiv.org/abs/2506.09985>
- [4] V-JEPA 2.1 (2026). <https://arxiv.org/abs/2603.14482>

### C. CLIP Papers

- [5] Learning Transferable Visual Models From Natural Language Supervision (CLIP) (2021). <https://arxiv.org/abs/2103.00020>
- [6] VideoCLIP: Contrastive Pre-training for Zero-shot Video-Text Understanding (2021). <https://arxiv.org/abs/2109.14084>
- [7] VadCLIP: Adapting CLIP for Video Anomaly Detection (2023). <https://arxiv.org/abs/2307.13425>
- [8] CLIP-ReID (2022). <https://arxiv.org/abs/2211.13977>
- [9] RegionCLIP (2022). <https://arxiv.org/abs/2112.09106>

### D. CCTV / Surveillance Papers

- [10] AssistPDA: Online Video Surveillance Assistant for Video Anomaly Prediction, Detection and Analysis (2025). <https://arxiv.org/abs/2503.21904>

- [11] Cerberus: Cascaded Vision-Language Models for Real-Time Video Anomaly Detection (2025). <https://arxiv.org/abs/2510.16290>
- [12] VERA: Verbalized Learning for Explainable Video Anomaly Detection (2025). <https://openaccess.thecvf.com/>
- [13] The Evolution of Video Anomaly Detection: From DNN to MLLM (2025). <https://arxiv.org/abs/2507.21649>

## **E. Video Retrieval & RAG Papers**

- [14] VideoRAG: Retrieval-Augmented Generation over Video Corpus (2025). <https://arxiv.org/abs/2501.05874>
- [15] VideoRAG for Extremely Long Context Videos (2025). <https://arxiv.org/abs/2502.01549>
- [16] MissionGNN (2025). <https://openaccess.thecvf.com/>
- [17] GenDFIR (2024). <https://arxiv.org/abs/2409.02572>

## **F. Detection, Tracking & Video Representation Papers**

- [18] VideoMAE (2022). <https://arxiv.org/abs/2203.12602>
- [19] SlowFast Networks for Video Recognition (2019). <https://arxiv.org/abs/1812.03982>
- [20] ByteTrack (2021). <https://arxiv.org/abs/2110.06864>
- [21] DeepSORT (2017). <https://arxiv.org/abs/1703.07402>
- [22] ST-GCN (2018). <https://arxiv.org/abs/1801.07455>

## **G. LLM Hallucination & Faithfulness Papers**

- [23] SelfCheckGPT (2023). <https://arxiv.org/abs/2303.08896>
- [24] Hallucination in Large Language Models: A Survey (2023). <https://arxiv.org/abs/2311.05232>
- [25] Faithfulness in Natural Language Generation (2022). <https://arxiv.org/abs/2207.07408>

## **H. Cross-Camera Generalization Papers**

- [26] CLIP-ReID (2022). <https://arxiv.org/abs/2211.13977>
- [27] TransReID (2021). <https://arxiv.org/abs/2102.04378>
- [28] Deep Learning for Person Re-identification: A Survey (2020). <https://arxiv.org/abs/2001.04193>